

GBU 2301: Business Statistics

Week 2: Tabular and Graphical Displays

Fall 2026 | Dr. Omer Cayirli | Assistant Professor of Finance

Required reading: Anderson et al., Statistics for Business and Economics, 15th ed., Chapter 2

Outline

- Summarizing One Variable
 - Categorical Data
 - Quantitative Data
- Summarizing Two Variables
 - Crosstabulations
 - Graphical Displays
- Data Visualization
 - Creating Effective Displays
 - Choosing a Display
 - Data Dashboards

Summarizing Categorical Data

- Frequency Distribution
- Relative Frequency Distribution
- Percent Frequency Distribution
- Bar Chart
- Pie Chart

Pelican Stores

The Pelican Stores data contain 100 customer records and eight variables.

Variable	Meaning
customer	Customer identifier
customer_type	Regular or promotional customer
items	Number of items purchased
net_sales	Net sales in dollars
payment_method	Method of payment
gender	Recorded customer gender
marital_status	Recorded marital status
age	Customer age in years

The data are supplied with Anderson et al. One row represents one recorded customer.

Pelican Stores

Observations: 100 | Variables: 8

	customer	customer_type	items	net_sales	payment_method	gender	marital_status	age
0	1	Regular	1	39.50	Discover	Male	Married	32
1	2	Promotional	1	102.40	Proprietary Card	Female	Married	36
2	3	Regular	1	22.50	Proprietary Card	Female	Married	32
3	4	Promotional	5	100.40	Proprietary Card	Female	Married	28
4	5	Regular	2	54.00	MasterCard	Female	Married	34
5	6	Regular	1	44.50	MasterCard	Female	Married	44
6	7	Promotional	2	78.00	Proprietary Card	Female	Married	30
7	8	Regular	1	22.50	Visa	Female	Married	40
8	9	Promotional	2	56.52	Proprietary Card	Female	Married	46
9	10	Regular	1	44.50	Proprietary Card	Female	Married	36

Frequency Distribution for Categorical Data

A frequency distribution lists the number of observations in each category.

The categories must be:

- Nonoverlapping: an observation is not counted in two categories.
- Collectively exhaustive: every observation belongs to a category.

The sum of the class frequencies must equal the number of observations, n .

When rare categories are combined into an Other category, the rule used to combine them should be stated.

Relative and Percent Frequency Distributions

For a class with frequency f and a data set containing n observations:

$$\text{Relative frequency} = \frac{f}{n}$$

$$\text{Percent frequency} = 100 \left(\frac{f}{n} \right)$$

The relative frequencies sum to 1, and the percent frequencies sum to 100 percent.

Counts describe volume. Relative and percent frequencies allow comparisons when group totals differ.

	payment_method	frequency	relative_frequency	percent_frequency
0	Proprietary Card	70	0.70	70.0
1	MasterCard	14	0.14	14.0
2	Visa	10	0.10	10.0
3	Discover	4	0.04	4.0
4	American Express	2	0.02	2.0

Frequency total: 100

Relative-frequency total: 1.0

Bar Chart

A bar chart represents the frequency, relative frequency, or percent frequency of categorical data.

- Each bar represents a distinct category.
- The bars are separated because adjacent categories do not form a numerical interval.
- Sorting bars can make rankings easier to see.
- A count scale and a percent scale answer different questions when group totals differ.

Because bar length carries the comparison, the numerical axis should generally begin at zero.

Pie Chart

A pie chart represents categories as parts of a whole. The size of each sector is determined by its relative or percent frequency.

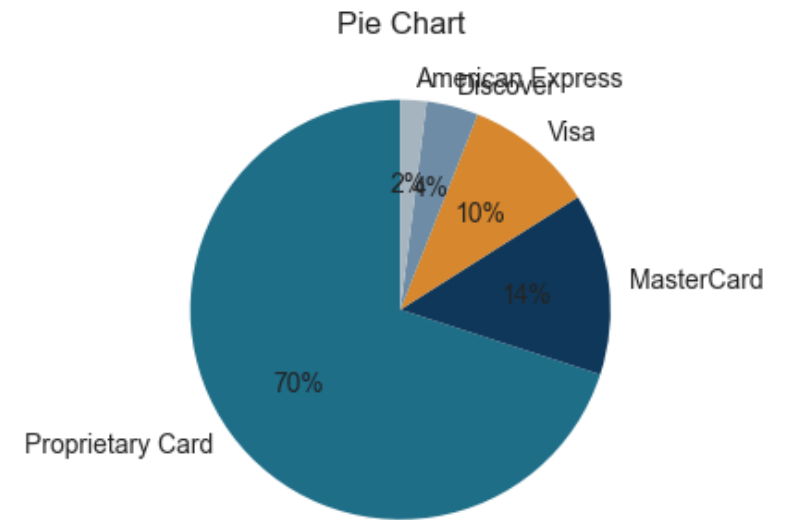
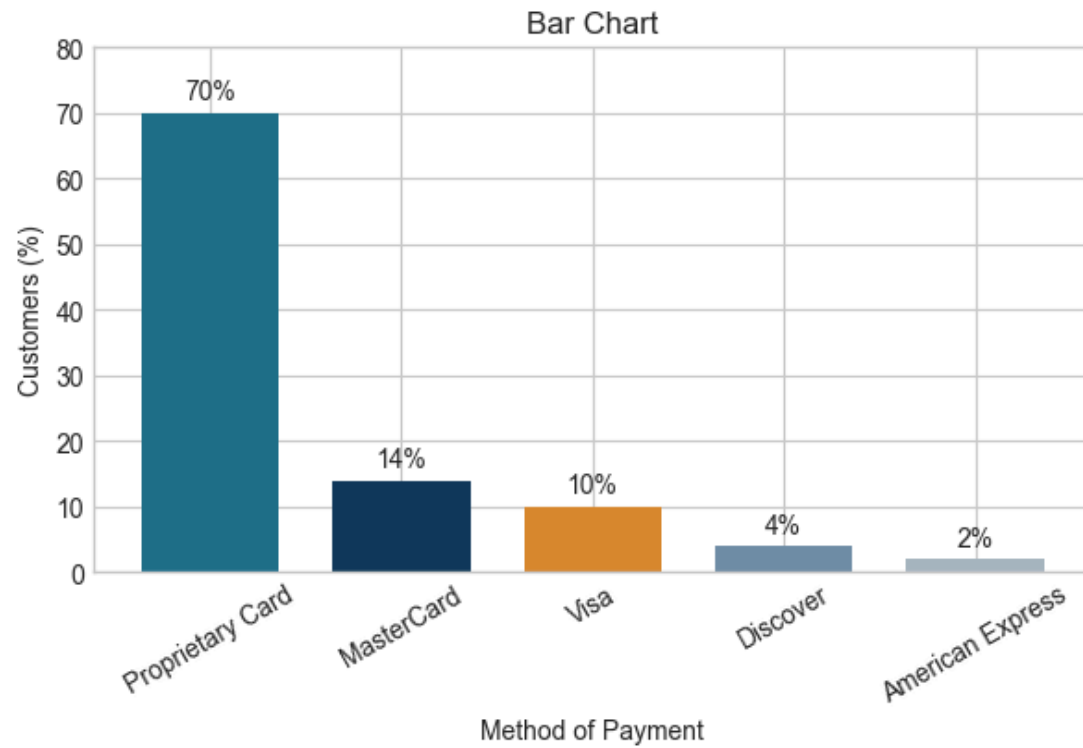
The two displays contain the same frequency information, but they do not support comparison equally well.

- Bar lengths share a common baseline and are easier to compare.
- Pie charts become difficult to read when there are many categories or similar shares.
- Three-dimensional effects distort sector sizes without adding information.

A pie chart may communicate a simple composition. A bar chart is generally the stronger default for comparison.

Pie Chart

Pelican Stores: Method of Payment



Exercise 1

Pelican Stores recorded four Discover customers and two American Express customers in the sample of 100. The finance manager proposes discontinuing both payment methods because their frequencies are small. The proposed report contains only the five frequency counts and a pie chart.

Prepare a recommendation for management. Determine how the payment-method data should be summarized and displayed, whether the two categories should be combined, what the sample supports, and what further evidence is needed before payment options are removed.

Exercise 1: Solution

- The report should include counts and percentages. A sorted bar chart makes the size difference clearer than a pie chart.
- Discover and American Express together account for six of the 100 recorded customers. Combining them into Other may simplify a display, but it would hide the distinction needed for a decision about individual payment methods.
- The sample describes the recorded customers. It does not establish current demand across stores, seasons, customer groups, or transaction values.
- A decision also requires processing costs, net sales by payment method, trends over time, declined transactions, and the value of customers who require each method.
- Management should retain the separate categories in the analytical table and postpone the decision until the operational and financial consequences are evaluated.

Frequency Distribution for Quantitative Data

A quantitative variable may contain many distinct values. Grouping values into nonoverlapping classes reveals concentration and variation that can be difficult to see in the raw rows.

Constructing a grouped frequency distribution requires decisions about:

1. The number of classes
2. The width of the classes
3. The class limits

The classes must cover the observed range, and each observation must belong to exactly one class. Relative and percent frequencies are calculated from the class counts in the same way as for categorical data.

Number of Classes and Class Width

An approximate class width is

$$\text{Approximate class width} = \frac{\text{largest value} - \text{smallest value}}{\text{number of classes}}$$

For Pelican Stores net sales:

$$\frac{287.59 - 13.23}{6} = 45.73$$

A convenient width of 50 dollars produces six classes. The number of classes and class width are judgment choices. Different choices can be equally defensible.

Class Limits and Class Midpoints

Class limits specify which observations belong to a class. For net sales measured to the nearest cent, one possible set of classes is:

- 0.00 to 49.99
- 50.00 to 99.99
- 100.00 to 149.99
- 150.00 to 199.99
- 200.00 to 249.99
- 250.00 to 299.99

The class midpoint is halfway between the lower and upper class limits. The first midpoint is approximately 25 dollars.

The interval convention must be stated clearly, especially when values can lie on a boundary.

Class Limits and Class Midpoints

	frequency	relative_frequency	percent_frequency	cumulative_frequency	cumulative_percent
net_sales_class					
0.00 to 49.99	39	0.39	39.0	39	39.0
50.00 to 99.99	35	0.35	35.0	74	74.0
100.00 to 149.99	16	0.16	16.0	90	90.0
150.00 to 199.99	6	0.06	6.0	96	96.0
200.00 to 249.99	1	0.01	1.0	97	97.0
250.00 to 299.99	3	0.03	3.0	100	100.0

Dot Plot

A dot plot places one dot above the horizontal axis for each observation. Repeated values are stacked.

A dot plot preserves the observed values and can reveal:

- Concentrations
- Gaps
- Repeated values
- Unusual observations

Dot plots are most readable for small or moderate data sets and variables with a manageable number of distinct values.

Dot Plot



Histogram

A histogram represents the frequency, relative frequency, or percent frequency of quantitative observations within class intervals.

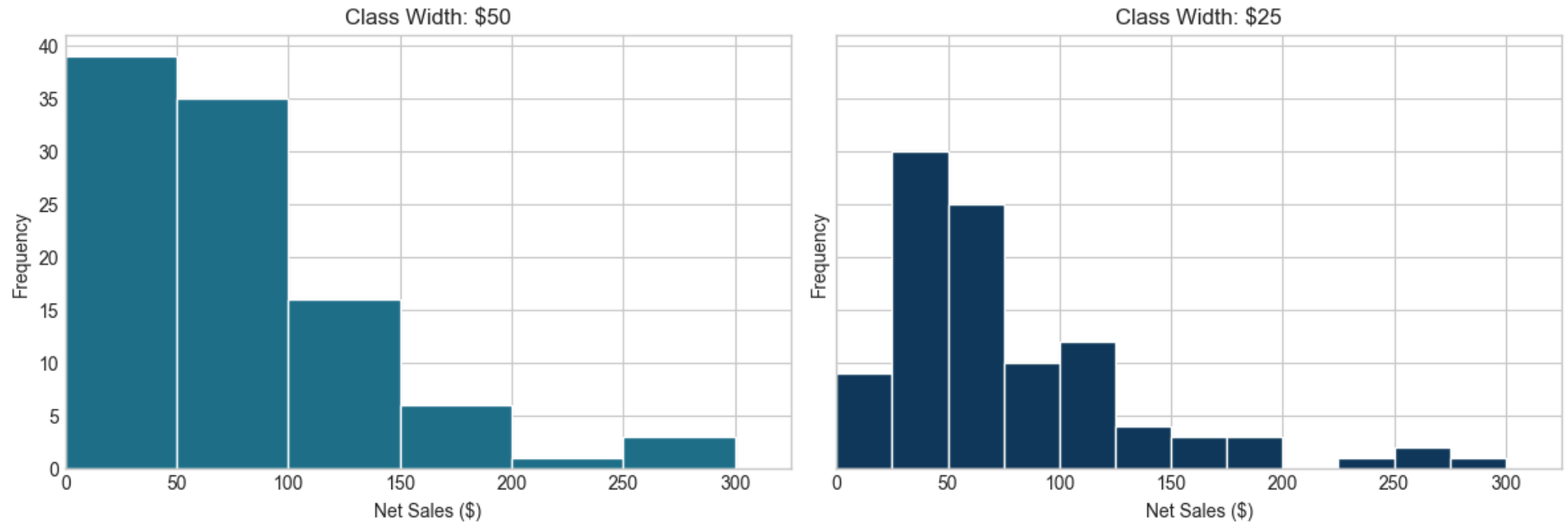
Histogram rectangles touch because adjacent classes represent consecutive numerical intervals. Their order cannot be rearranged.

A bar chart represents separate categories, normally with gaps between bars. The categories may often be reordered without changing their meaning.

The visual similarity between the displays does not make them interchangeable.

Histogram

Pelican Stores: Two Histograms of the Same Data



Distribution Shape and Class Choice

A distribution may be approximately symmetric, skewed to the left, or skewed to the right. The tail points in the direction of the skew.

Pelican Stores net sales are right-skewed. Most recorded sales are below 100 dollars, while a small number extend beyond 200 dollars.

The broad shape remains visible in both histograms, but the apparent peaks, gaps, and local concentrations change with the class width.

A responsible interpretation distinguishes conclusions that are stable across reasonable class choices from features created by one particular grouping.

Cumulative Distributions

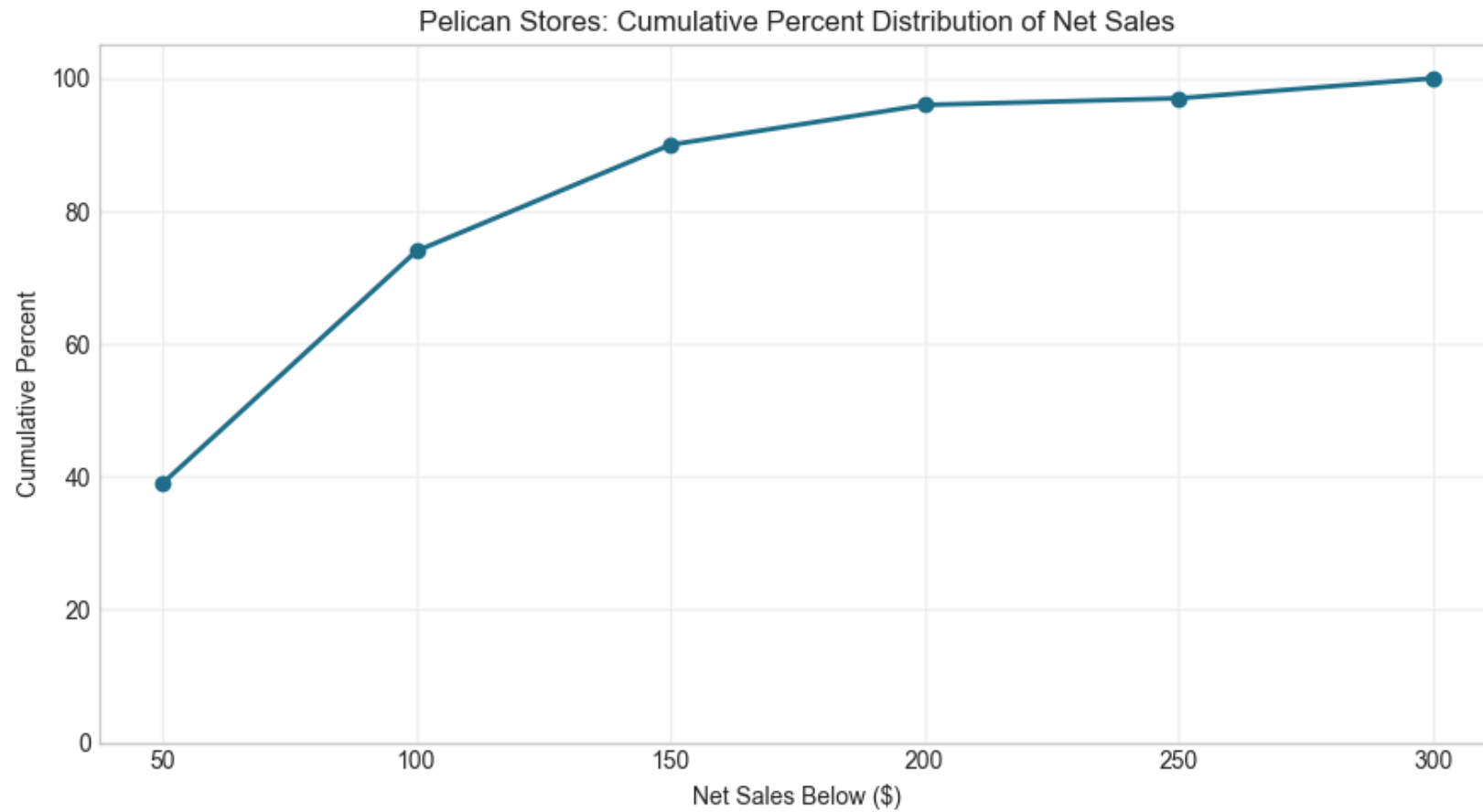
A cumulative distribution adds the classes from the lowest class through a stated upper class limit.

- Cumulative frequency: number of observations less than or equal to the upper limit
- Cumulative relative frequency: corresponding proportion
- Cumulative percent frequency: corresponding percentage

From the grouped Pelican Stores table, 74 customers have net sales below 100 dollars and 96 have net sales below 200 dollars.

Cumulative summaries answer threshold questions. They do not show the frequency inside a single class unless adjacent cumulative values are compared.

Cumulative Distributions



Stem-and-Leaf Display

A stem-and-leaf display shows both rank order and distribution shape while retaining the observed values. For the ages of the first 30 Pelican Stores customers:

Leaf unit = 1 year

```
2 | 2 4 8 8
3 | 0 2 2 2 2 4 6 6 6 8
4 | 0 0 0 0 2 2 4 6 6 8 8
5 | 0 4 6
6 | 0
7 | 0
```

The value 4 | 6 represents age 46. A crowded stem may be stretched into separate rows for leaves 0 to 4 and 5 to 9. If the leaf unit is 10, 0.1, or another value, it must be reported.

Exercise 2

Two analysts use the Pelican Stores net-sales data to support a proposed customer segmentation. One uses class widths of 50 dollars and describes a broad low-sales group with a long upper tail. The other uses class widths of 25 dollars and claims that the additional peaks identify several distinct customer groups.

Management asks which interpretation should guide the segmentation decision. Evaluate the two analyses, identify conclusions that are supported regardless of the class width, and recommend what should be examined before the visible groups are treated as genuine customer segments.

Exercise 2: Solution

- Both class widths are defensible summaries. A histogram does not have a uniquely correct number of classes.
- Stable conclusions include that 74 of 100 observations are below 100 dollars, only four are at least 200 dollars, and the distribution has a long right tail.
- The local peaks change when the width changes. They are not sufficient evidence of distinct customer groups.
- The analysts should inspect the raw values, a dot plot or cumulative distribution, and histograms using other reasonable boundaries. Business-relevant thresholds should be justified independently of the most visually convenient result.
- A segmentation decision also requires evidence that the proposed groups differ meaningfully on behavior, costs, or future value. This cross-sectional sales variable alone does not provide that evidence.

Crosstabulation

A crosstabulation summarizes two variables. The classes of one variable form the rows, and the classes of the other form the columns.

The cells show joint frequencies. The margins show the frequency distribution of each variable separately.

Crosstabulations can summarize:

- Two categorical variables
- One categorical variable and a quantitative variable divided into classes
- Two quantitative variables after both are divided into classes

Grouping quantitative values loses detail, so the classes must serve the analytical question.

Row and Column Percentages

Counts alone can be misleading when row or column totals differ. Conditional percentages use a denominator selected by the question.

For a table with customer type in the rows and payment method in the columns:

- Row percentages answer: Within each customer type, how are payment methods distributed?
- Column percentages answer: Within each payment method, how are customer types distributed?

The same cell therefore has two possible conditional percentages. Neither is automatically correct. The comparison determines the denominator.

Row and Column Percentages

Frequencies

payment_method	Proprietary Card	MasterCard	Visa	Discover	American Express
customer_type					
Promotional	57	7	5	0	1
Regular	13	7	5	4	1

Row percentages

payment_method	Proprietary Card	MasterCard	Visa	Discover	American Express
customer_type					
Promotional	81.4	10.0	7.1	0.0	1.4
Regular	43.3	23.3	16.7	13.3	3.3

Column percentages

payment_method	Proprietary Card	MasterCard	Visa	Discover	American Express
customer_type					
Promotional	81.4	50.0	50.0	0.0	50.0
Regular	18.6	50.0	50.0	100.0	50.0

Simpson's Paradox

An aggregate comparison can reverse after the data are separated into relevant groups. This is Simpson's paradox.

The reversal occurs because the groups can have different baseline outcomes and different representation in the alternatives being compared.

A sound analysis therefore asks:

- Which groups were combined?
- Are their baseline outcomes different?
- Are the alternatives represented in those groups in different proportions?
- Which conditional comparison answers the decision question?

Aggregation is not automatically wrong. Its interpretation depends on the structure hidden inside the total.

Exercise 3

A retailer tests two campaigns. Management observes an overall conversion rate of 26.4 percent for Campaign A and 73.6 percent for Campaign B and plans to move the full budget to Campaign B.

Customer segment	Campaign A	Campaign B
High intent	9 conversions from 10 customers	80 conversions from 100 customers
Low intent	20 conversions from 100 customers	1 conversion from 10 customers
Overall	29 conversions from 110 customers	81 conversions from 110 customers

Prepare a recommendation on the proposed budget decision using the complete table.

Exercise 3: Solution

- Within the high-intent segment, Campaign A converts 90 percent and Campaign B converts 80 percent. Within the low-intent segment, Campaign A converts 20 percent and Campaign B converts 10 percent.
- Campaign A has the higher conversion rate in both segments, although Campaign B has the higher aggregate rate. The aggregate comparison reverses the conditional comparisons.
- Campaign A was shown mostly to low-intent customers, while Campaign B was shown mostly to high-intent customers. The customer mix, rather than campaign superiority, explains the aggregate ranking.
- Management should not shift the budget on the basis of the aggregate rate. The campaigns should be compared within relevant segments, with customer assignment and sample sizes reported.
- A causal budget decision requires a credible assignment process, preferably random assignment within customer segments.

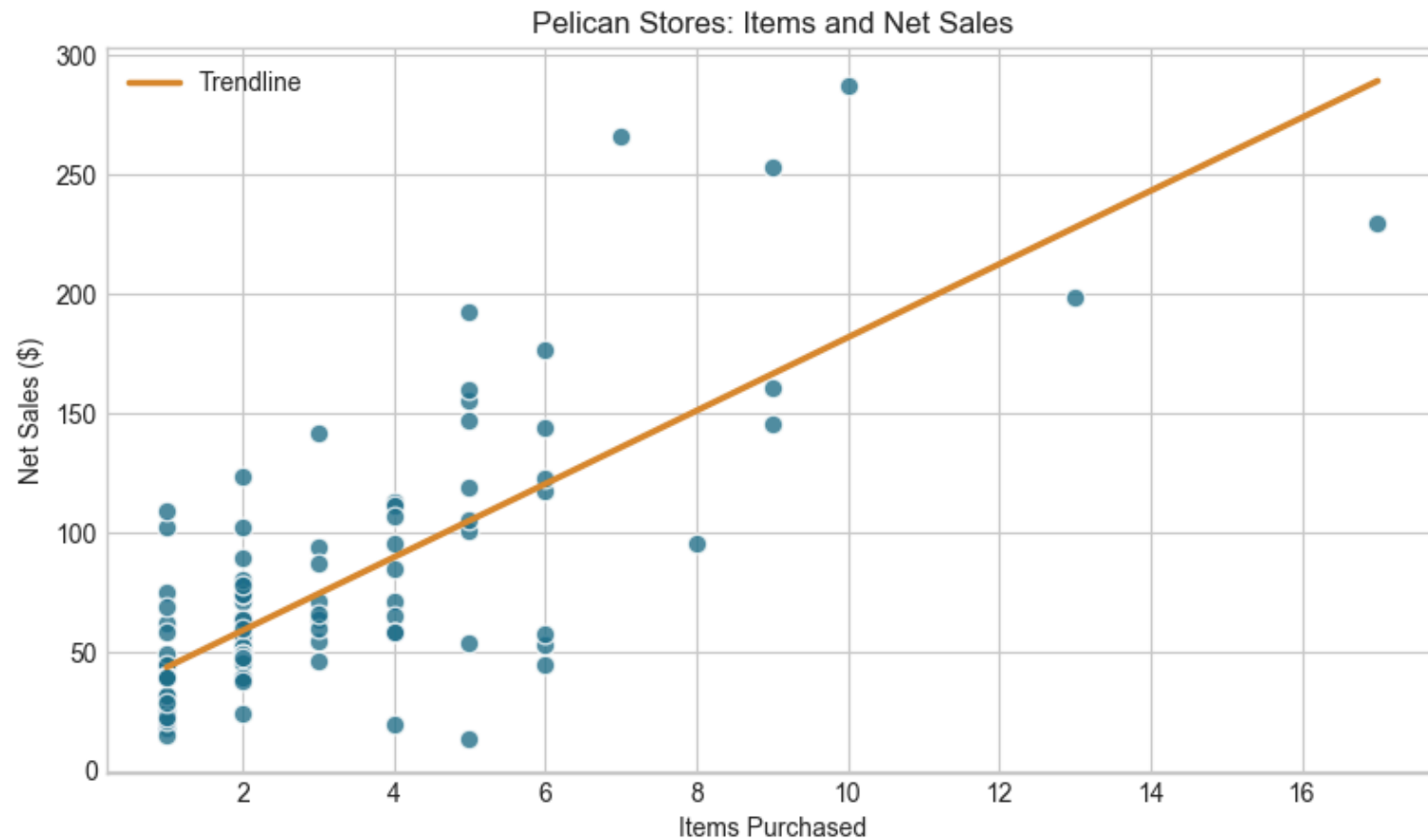
Scatter Diagram and Trendline

A scatter diagram displays the relationship between two quantitative variables. Each observation contributes one point.

- The explanatory or organizing variable is usually placed on the horizontal axis.
- The response or outcome of interest is usually placed on the vertical axis.
- The pattern of points can reveal direction, form, strength, clusters, and unusual observations.

A trendline provides an approximation of the overall relationship. It does not make the relationship causal and does not eliminate variation around the line.

Scatter Diagram and Trendline



Reading a Scatter Diagram

The Pelican Stores display shows a positive association: customers purchasing more items tend to record higher net sales.

- The relationship is not exact.
 - Customers purchasing the same number of items can have different net sales because item prices and discounts differ.
 - Several observations also lie far from the main concentration.
- A positive pattern does not establish that increasing item count alone will produce the increase represented by the trendline.
 - The graph describes association in the recorded data.
- No apparent relationship is also a meaningful result.
 - It means that the display does not reveal a systematic pattern, not that the variables are proven unrelated in every population or under every form of relationship.

Side-by-Side and Stacked Bar Charts

Side-by-side bars place category values next to one another. They support direct comparison across groups.

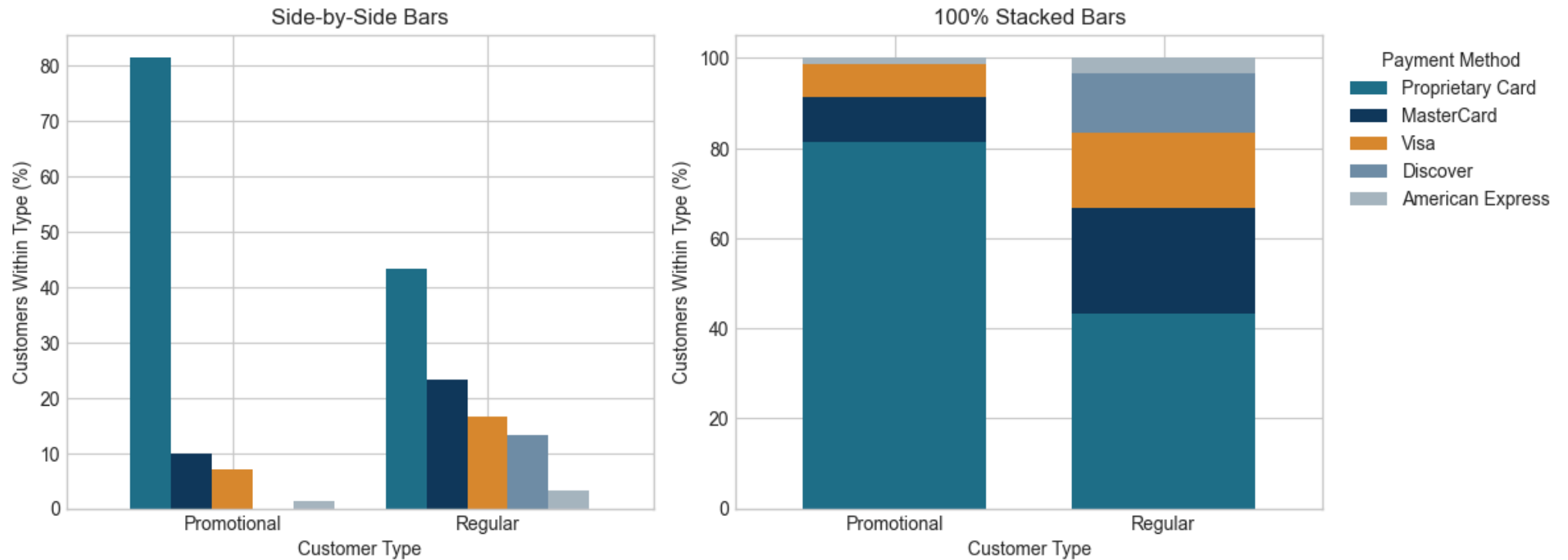
Stacked bars divide each bar into segments. They emphasize how a total is composed.

- Stacked counts preserve differences in group size.
- A 100 percent stacked chart gives every group the same total length and compares composition.
- Internal stacked segments do not share a common baseline and can be difficult to compare.

The choice between counts and percentages depends on whether the question concerns volume or composition.

Side-by-Side and Stacked Bar Charts

Pelican Stores: Payment Method by Customer Type



Exercise 4

Pelican Stores management makes two claims:

1. Increasing the number of items in a basket will produce a predictable increase in net sales.
2. Promotional customers are more likely than regular customers to adopt the proprietary card.

Management wants one graphical report to support both claims. Recommend the displays and comparisons that should be used, assess what the observed data support, and identify the parts of the claims that go beyond the evidence.

Exercise 4: Solution

- The first claim concerns two quantitative variables. A scatter diagram of items and net sales with a trendline shows a positive but imperfect association. Variation at the same item count prevents a predictable one-for-one conclusion.
- The word produce is causal. The cross-sectional display does not show that deliberately increasing basket size would cause the trendline increase. Item prices, product mix, and discounts also matter.
- The second claim concerns two categorical variables. Row percentages compare payment-method composition within promotional and regular customers. Side-by-side or 100 percent stacked bars can display those conditional percentages.
- The proprietary-card share is higher among promotional customers in these data, but the sample does not establish that promotion caused card adoption.
- The report needs two displays because the claims involve different variable structures and different comparisons.

Creating Effective Graphical Displays

An effective display communicates the relevant information clearly and accurately.

- Use a clear title, axis labels, and units.
- Keep the display simple and avoid unnecessary dimensions or decoration.
- Use distinct colors only when they carry information.
- Place a necessary legend close to the data.
- Report the population, period, denominator, and source when they affect interpretation.
- Use a zero baseline for bar lengths and avoid scales that exaggerate small differences.

A technically correct chart can still mislead through omitted categories, unsuitable denominators, distorted scales, or selective time periods.

Choosing a Graphical Display

Purpose	Variable structure	Useful displays
Show a categorical distribution	One categorical variable	Bar chart; pie chart for a simple composition
Show a quantitative distribution	One quantitative variable	Dot plot, histogram, stem-and-leaf display
Compare categorical distributions	Two categorical variables	Side-by-side or 100 percent stacked bars
Show composition and volume	Two categorical variables	Stacked counts with totals reported
Show a quantitative relationship	Two quantitative variables	Scatter diagram and, when useful, a trendline
Show change through time	Quantitative variable indexed by time	Time-series plot

The most elaborate display is rarely the most informative. The display should make the intended comparison easy to see.

Data Dashboards

A data dashboard is a coordinated set of visual displays used to monitor the performance of an organization or process.

A useful dashboard identifies:

- The decision maker and the decisions to be supported
- A limited set of relevant key performance indicators
- Current performance, comparison values, and targets
- The required update frequency
- Exceptions that require attention

A page containing several charts is not automatically a dashboard. The displays must work together, use consistent definitions, and inform recurring decisions without overwhelming the user.

Exercise 5

A store director calls a static page based on the 100 Pelican Stores observations a performance dashboard. It contains a three-dimensional pie chart of payment methods, a net-sales histogram with no units, a customer-type bar chart whose numerical axis begins at 60 percent, and a scatter diagram of items and net sales. The page provides no reporting period, targets, comparison values, or update schedule.

Prepare a recommendation on whether the page can support ongoing store management and how it should be redesigned.

Exercise 5: Solution

- The page is a collection of descriptive displays from one sample, not an operational dashboard. It does not identify recurring decisions, current performance, targets, or time comparisons.
- Replace the three-dimensional pie chart with a sorted bar chart when payment-method composition is relevant. Label the histogram in dollars and report its class definition. Begin the customer-type bar chart at zero.
- Retain the scatter diagram only if the items-sales relationship informs a defined decision. It should not be included merely because it is available.
- Define the intended user, reporting period, update frequency, KPIs, comparison values, and action thresholds before choosing charts.
- A genuine dashboard would require regularly updated operational data. The 100-record sample can illustrate possible displays but cannot monitor current store performance.

What is next?

- Descriptive Statistics II
 - Numerical Measures
 - Reading(s):
 - SBE: Ch. 3

Exercise 6

Northstar Pharmacies ranks 12 branches by the number of customer complaints and plans to sanction the three branches with the highest counts.

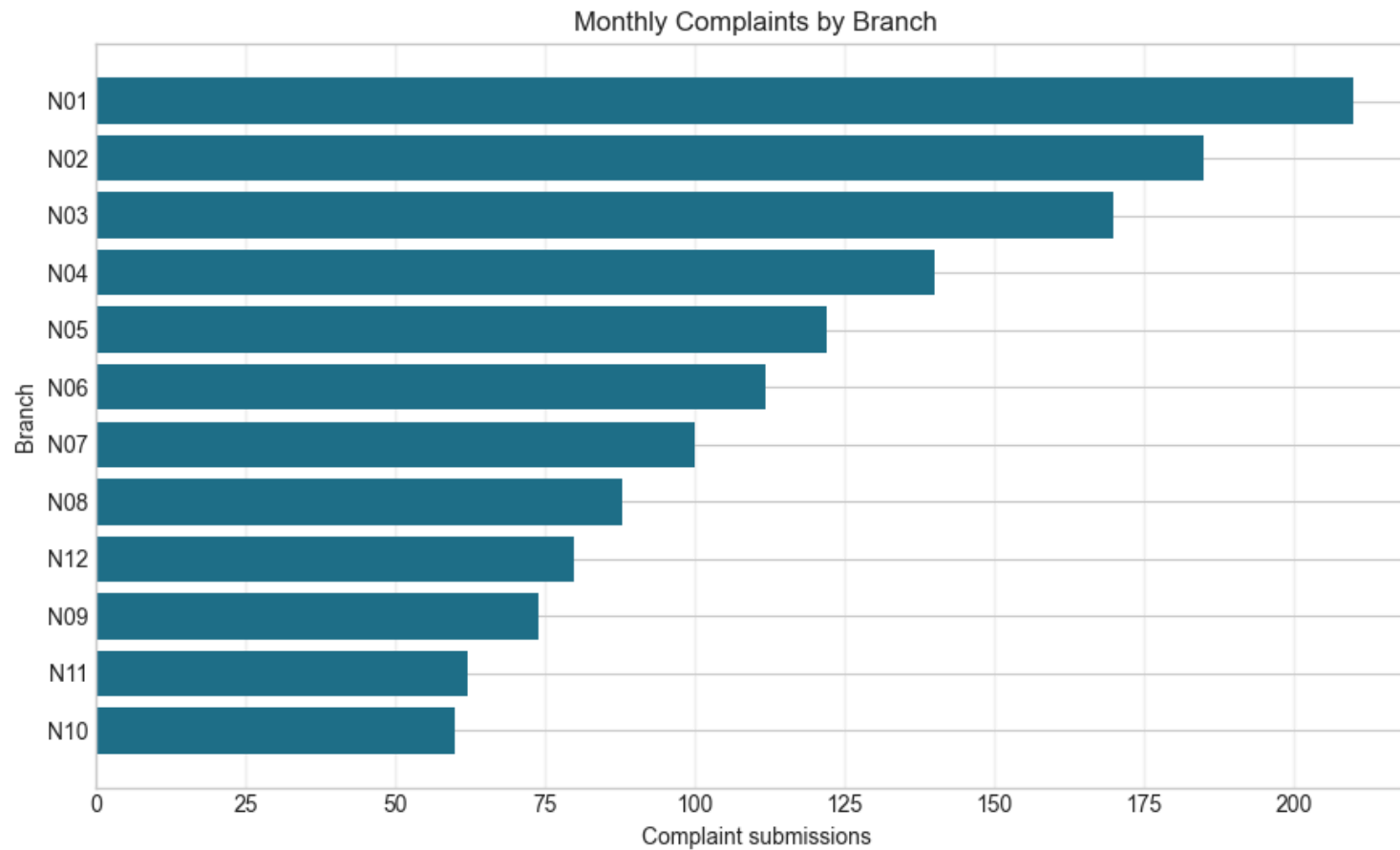
- Monthly transactions range from 8,000 to 55,000 across branches.
- Complaints are classified as billing, delivery, product availability, or service.
- A customer can submit more than one complaint about the same transaction.

Audit the data and prepare a recommendation for the operations director.

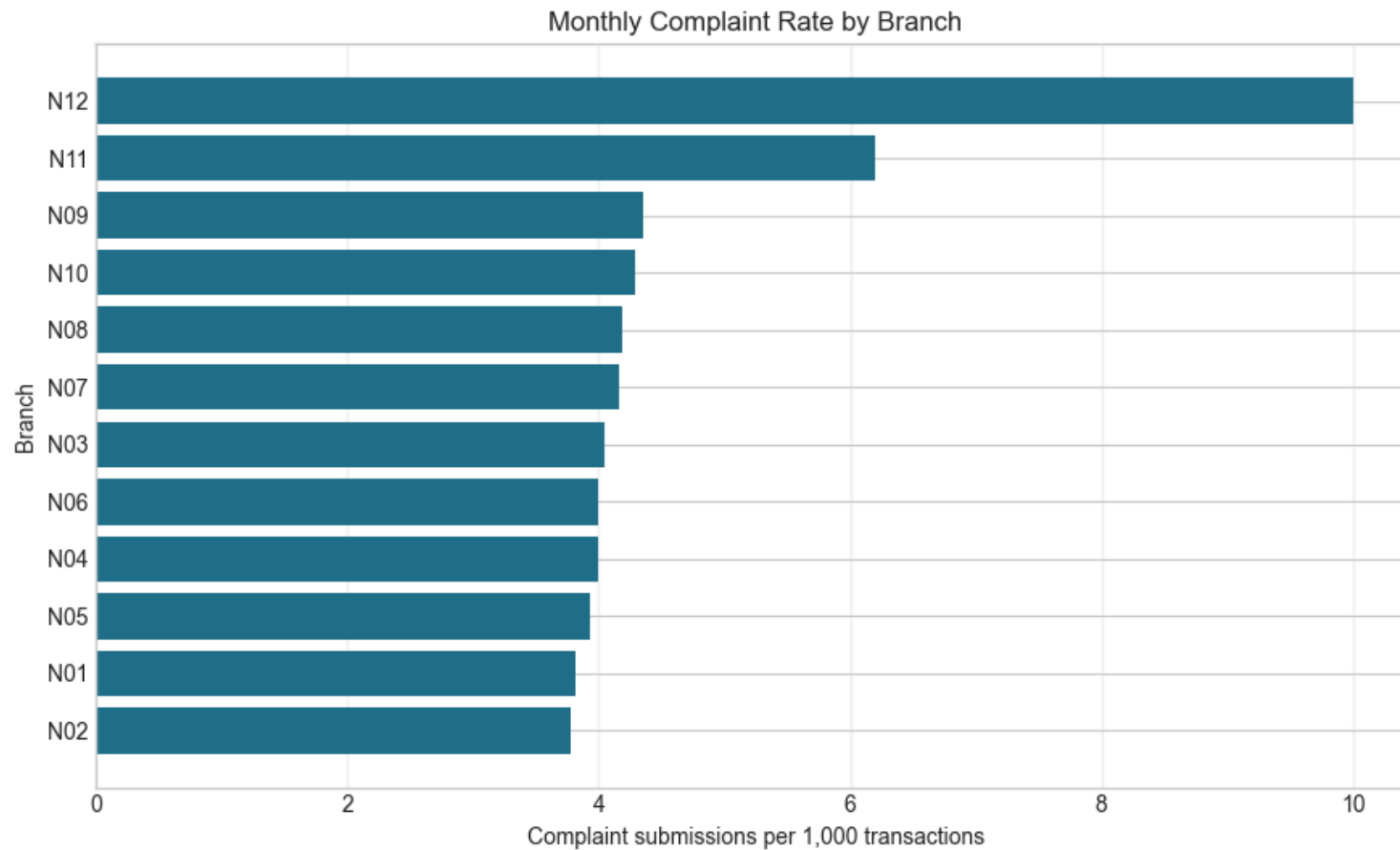
Exercise 6

	branch	transactions	complaints	billing	delivery	availability	service
0	N01	55000	210	20	35	95	60
1	N02	49000	185	75	40	30	40
2	N03	42000	170	32	60	38	40
3	N04	35000	140	20	30	55	35
4	N05	31000	122	17	30	44	31
5	N06	28000	112	18	26	39	29
6	N07	24000	100	15	26	35	24
7	N08	21000	88	14	22	29	23
8	N09	17000	74	13	18	24	19
9	N10	14000	60	9	16	21	14
10	N11	10000	62	11	16	20	15
11	N12	8000	80	14	20	25	21

Exercise 6



Exercise 6



Exercise 6: Solution

- Complaint count measures volume, but branches have very different transaction exposure. Counts alone do not provide a fair comparison of complaint incidence.
- The element must be defined.
 - Complaint events, complaining customers, and transactions with at least one complaint answer different questions.
 - Repeated complaints about one transaction must not be treated as independent affected transactions.
- Report branch transaction totals and complaint rates using a clearly defined denominator.

Exercise 6: Solution

- A sorted bar chart of complaint rates supports branch comparison.
 - A separate volume chart or labels can preserve operational scale.
- A 100 percent stacked bar chart can display complaint-category composition across branches.
 - Counts should accompany percentages when some branches have few complaints.
- Before sanctions, examine reporting consistency, transaction mix, complaint severity, time patterns, and whether customers have equal access to the complaint channel.
- The report can identify branches for investigation. It does not by itself establish poor management or the cause of the complaints.

Exercise 7

A bank is testing two online loan application interfaces. A manager recommends Interface B because more completed applications finished in under five minutes. The table includes completed applications only.

Completion time (minutes)	Interface A	Interface B
0 to under 5	50	80
5 to under 10	90	40
10 to under 15	45	20
15 to under 20	15	60

Each interface has 200 completed applications. Abandoned applications are not included.

Assess the manager's recommendation using the full distribution. What should management see before deciding whether to replace A?

Exercise 7: Solution

B has more completions under five minutes: $80/200 = 40$ percent, compared with $50/200 = 25$ percent for A. But A has more completions under ten minutes: $140/200 = 70$ percent, compared with $120/200 = 60$ percent for B. At the slow end, $60/200 = 30$ percent of B completions take 15 to under 20 minutes, compared with $15/200 = 7.5$ percent for A. B is not consistently faster across the distribution.

Use histograms with the same time classes and a cumulative-percent display to show the tradeoff. Do not replace A on the basis of completed applications alone. Record how many applications were started and abandoned under each interface, then compare completion and time for comparable applicants before rollout.

Exercise 8

An online retailer reports that a four-week email campaign is succeeding because weekly net sales rose from 42,000 to 53,900. Management wants to expand the campaign. It ran throughout the four weeks.

Week	Site visits	Purchases	Net sales (\$)
1	2,000	600	42,000
2	2,500	750	52,500
3	3,000	780	54,600
4	3,500	770	53,900

Prepare a recommendation. What does the four-week pattern actually show, and what should management monitor before expanding the campaign?

Exercise 8: Solution

Purchases per visit fall from 30 percent in weeks 1 and 2 to 26 percent in week 3 and 22 percent in week 4. Net sales rise from week 1 to week 4, but both purchases and sales peak in week 3. More visits are not translating into proportionally more purchases.

Show visits, purchases, net sales, and purchases per visit by week, using separate labeled scales for counts, dollars, and percentages. The four campaign weeks have no pre-campaign baseline or comparison group, so the data cannot establish that the campaign caused the sales increase. Do not expand on this report alone; compare with a suitable noncampaign group and include campaign costs before deciding.